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Lab:2 - Cloud Security trends

Step:1

Microsoft Azure

Search resources, services, and docs (G+)

Copilot

pate1562@algonquiniv...
ALGONQUIN COLLEGE (ALGONQ...

[Home](#) > [Allowed locations](#) >

Assign policy

Basics

Parameters

Remediation

Managed identity

Non-compliance messages

Review + create

Scope

Scope *

Azure for Students

Learn more about setting the scope

Exclusions

Optionally select resources to exclude from the policy assignment.

Resource selectors

Expand

Using resource selectors, you can further refine this assignment's applicability by targeting specific subsets of resources. Expand to learn more.

Basics

Policy definition *

Allowed locations

Version *

1.0.*

Overrides

Expand

Using overrides, you can change the effects or referenced versions of definitions for all or a subset of resources evaluated by this assignment. Expand to learn more.

Assignment name *

Allowed locations

Description

Policy enforcement

Enabled

Previous

Next

Review + create

Give feedback

Microsoft Azure

Search resources, services, and docs (G+)

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ALGONQUIN COLLEGE (P...

[Home](#) > [Policy | Assignments](#) >

Allowed locations

Policy Assignment

[Edit assignment](#) [Delete assignment](#) [Duplicate assignment](#) [View compliance](#) [View definition](#) [Create exemption](#) [Create remediation task](#)

Essentials

Name

: Allowed locations

Scope

: Azure for Students

Definition version

: 1.0.*

Excluded scopes

: --

Description

: --

Definition type

: Policy

Assignment ID

: /subscriptions/69be1f0a-ef3f-4038-96d6-3f6d028fd393/providers/microsoft.authorization/po...

Policy enforcement

: Default

Assigned by

: --

Parameters (1)

Resource selectors (0)

Overrides (0)

Exemptions (0)

Remediation (0)

Deployed resources

Managed Identity

Search by parameter name

All types

Parameter ID

Parameter name

Parameter value

Policy assignment parameter reference type

listOfAllowedLocations

Allowed locations

["eastus2"]

User defined parameter

Edit columns

Microsoft Azure

Search resources, services, and docs (G+/I)

Copilot

Home > Policy | Assignments > Allowed locations >

Assign policy

BasicsParametersRemediationManaged identityNon-compliance messagesReview + create

Basics

Scope

Azure for Students

Exclusions

--

Policy definition

Allowed locations

Assignment name

Allowed locations

Version

1.0.*

Description

--

Policy enforcement

Default

Assigned by

Arpitkumar Patel

Advanced

Resource selectors

No selectors associated with this assignment.

Overrides

No overrides associated with this assignment.

Parameters

Allowed locations

["eastus2"]

Remediation

Create a remediation task

No

Previous

Next

Save

Microsoft Azure

Search resources, services, and docs (G+/I)

Copilot

Home > Policy

Policy | Assignments

Assign policyAssign initiativeRefresh

Search

Filter by name or ID...

Scope : 2 selected

Definition type : All definition types

Total Assignments

3

Initiative Assignments

1

Policy Assignments

2

Edit columns

Export to CSV

Assignment name ↑	Scope ↑	Type ↑
Allowed resource deployment regions	Azure for Students	Policy
ASC Default (subscription: 69be1f0a-ef3f-4038-96d6-3f6d028fd393)	Azure for Students	Initiative
Allowed locations	Azure for Students	Policy

Microsoft Azure

Search resources, services, and docs (G+/I)

Copilot

Home > Storage center | Blob Storage >

Create a storage account

BasicsAdvancedNetworkingData protectionEncryptionTagsReview + create

Azure Storage is a Microsoft-managed service providing cloud storage that is highly available, secure, durable, scalable, and redundant. Azure Storage includes Azure Blobs (objects), Azure Data Lake Storage Gen2, Azure Files, Azure Queues, and Azure Tables. The cost of your storage account depends on the usage and the options you choose below. [Learn more about Azure storage accounts](#)

Project details

Select the subscription in which to create the new storage account. Choose a new or existing resource group to organize and manage your storage account together with other resources.

Subscription *

Azure for Students

Resource group *

(New) policy-test-rg

Create new

Instance details

Storage account name *

policytestlab2

Region *

(US) East US

Deploy to an Azure Extended Zone

This policy maintains a set of best available regions where your subscription can deploy resources. The objective of this policy is to ensure that your subscription has full access to Azure services with optimal performance. Should you need additional or different regions, contact support.

Preferred storage type

Choose preferred storage type

Previous

Next

Review + create

This helps us provide relevant guidance. It doesn't restrict your storage to this

Give feedback

Microsoft Azure

Search resources, services, and docs (G+/)

Copilot

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ALGONQUIN COLLEGE (ALGONQ...

Home > Storage center | Blob Storage >

Create a storage account

Validation failed. Required information is missing or not valid.

BasicsAdvancedNetworkingData protectionEncryptionTagsReview + create

Basics

Subscription	Azure for Students
Resource group	policy-test-rg
Location	East US
Storage account name	policytestlab2
Preferred storage type	
Performance	Standard
Replication	Read-access geo-redundant storage (RA-GRS)

Advanced

Enable hierarchical namespace	Disabled
Enable SFTP	Disabled
Enable network file system v3	Disabled
Allow cross-tenant replication	Disabled
Access tier	Hot
Enable large file shares	Enabled
Managed Identity for SMB	Disabled

Security

Previous

Next

Create

Give feedback

Explanation:

In this step, an Azure Policy named Allowed locations was assigned at the subscription level to restrict resource deployment to approved regions only. The policy was configured to allow specific regions, and an attempt was made to create a resource outside the permitted location (East US). Azure correctly blocked the deployment, confirming that the policy enforcement was working as intended.

Step:2

Microsoft Azure

Search resources, services, and docs (G+/)

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ALGONQUIN COLLEGE (ALGONQ...

Home > Network foundation | Virtual networks >

Create virtual network

Validation passed

BasicsSecurityIP addressesTagsReview + create

View automation template

Basics

Subscription	Azure for Students
Resource Group	policy-test-rg
Name	policy-lab
Region	East US 2

Security

Azure Bastion	Disabled
Azure Firewall	Disabled
Azure DDoS Network Protection	Disabled

IP addresses

Address space	10.0.0.0/16 (65,536 addresses)
Subnet	default (10.0.0.0/24) (256 addresses)

Tags

Previous

Next

Create

Download a template for automation

Microsoft Azure

Search resources, services, and docs (G+/)

Copilot

Home >

policy-lab-1768953587478 | Overview

Deployment

Search

Delete Cancel Redeploy Download Refresh

Overview

Inputs

Outputs

Template

✓ Your deployment is complete

Deployment name : policy-lab-1768953587478 Start time : 20/1/2026, 6:59:50 pm
Subscription : Azure for Students Correlation ID : 01f271f1-b6eb-4b79-80e4...
Resource group : policy-test-rg

> Deployment details

✓ Next steps

Go to resource

Cost management

Get notified to stay within your budget and prevent unexpected charges on your bill.
[Set up cost alerts >](#)

Microsoft Defender for Cloud

Secure your apps and infrastructure
[Go to Microsoft Defender for Cloud >](#)

Free Microsoft tutorials

[Start learning today >](#)

Work with an expert

Azure experts are service provider partners who can help manage your assets on Azure and be your first line of support.
[Find an Azure expert >](#)

Notifications

More events in the activity log →

Dismiss

✓ Deployment succeeded

Deployment 'policy-lab-1768953587478' to resource group 'policy-rg' was successful.

Go to resource Go to resource group

Microsoft Azure

Search resources, services, and docs (G+/)

Copilot

Home >

policy-lab

Virtual network

Diagnose issues with this virtual network Analyze connectivity issues in this virtual network Suggest connectivity model for this network

Search

Move Delete Refresh Give feedback

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Resource visualizer

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Help

Essentials

Resource group (move) : policy-test-rg

Location (move) : East US 2

Subscription (move) : Azure for Students

Subscription ID : 69be1f0a-ef3f-4038-96d6-3f6d028fd393

Address space : 10.0.0.0/16

Subnets : 1 subnet

DNS servers : Azure provided DNS service

BGP community string : Configure

Virtual network ID : 262ced61-33c2-4fd0-9075-2d63562ae90a

Tags (edit) : Add tags

Topology Properties Capabilities (5) Recommendations Tutorials

DDoS protection

Configure additional protection from distributed denial of service attacks.

Not configured

Azure Firewall

Protect your network with a stateful L3-L7 firewall.

Not configured

Peerings

Seamlessly connect two or more virtual networks.

Not configured

Microsoft Defender for Cloud

Strengthen the security posture of your environment.

Private endpoints

Privately access Azure services without sending traffic across internet.

Not configured

Explanation:

In this step, a virtual network named policy-lab was created to establish a secure network boundary for the lab environment. The virtual network was deployed in the East US 2 region under the existing resource group policy-test-rg. An address space of 10.0.0.0/16 was configured to allow sufficient IP capacity for future subnets. After reviewing the settings, the deployment was validated and successfully completed, confirming that the virtual network was created correctly.

Step:3

Microsoft Azure

Search resources, services, and docs (G+)

Copilot

Home > policy-lab

policy-lab | Subnets

Search

+ Subnet Refresh Manage users Delete Export to CSV

Create subnets to segment the virtual network address space into smaller ranges for use by your applications. When you deploy resources into a subnet, Azure assigns the resource an IP address from the subnet.

Search subnets

	Name ↑	IPv4	IPv6	Available IPs	Delegated to	Security group	Route table
☐	default	10.0.0.0/24	-	251	-	-	-
☐	private-sub...	10.0.1.0/24	-	251	-	-	-
☐	public-sub...	10.0.2.0/24	-	251	-	-	-

Showing 3 subnets

Subnets

Bastion

DDoS protection

Firewall

Microsoft Defender for Cloud

Network manager

DNS

Peerings

Service endpoints

Private endpoints

Properties

Locks

Monitoring

Automation

Help

Notifications

More events in the activity log → Dismiss all

Successfully added subnet

Successfully added subnet 'public-subnet' to virtual network 'policy-lab'.

a few seconds ago

Successfully added subnet

Successfully added subnet 'private-subnet' to virtual network 'policy-lab'.

2 minutes ago

Deployment succeeded

Deployment 'policy-lab-1768953587478' to resource group 'policy-test-rg' was successful.

Go to resource Go to resource group

20 minutes ago

Explanation:

In this step, the virtual network policy-lab was segmented into separate subnets to isolate resources based on access requirements. A private-subnet (10.0.1.0/24) was created to host internal resources with restricted access, while a public-subnet (10.0.2.0/24) was created for resources that require controlled external connectivity.

Step:4

Microsoft Azure

Search resources, services, and docs (G+)

Copilot

Home >

nsg-private

Network security group

Search

→ Move Delete Refresh Give feedback

How do I create an alert to track firewall metric failures Diagnose connectivity issues related to this security group Analyze security rules for this network security group

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Resource visualizer

Settings

Monitoring

Automation

Help

Essentials

Resource group (move) : policy-test-rg

Location : East US 2

Subscription (move) : Azure for Students

Subscription ID : 69be1f0a-ef3f-4038-96d6-3fd028fd393

Tags (edit) : Add tags

Custom security rules : 0 inbound, 0 outbound

Associated with : 0 subnets, 0 network interfaces

Filter by name

Port == all Protocol == all Source == all Destination == all Action == all

Priority ↑↓	Name ↑↓	Port ↑↓	Protocol ↑↓	Source ↑↓	Destination ↑↓	Action ↑↓
Inbound Security Rules						
65000	AllowVnetInBound	Any	Any	VirtualNetwork	VirtualNetwork	Allow
65001	AllowAzureLoadBalancerInBou...	Any	Any	AzureLoadBalancer	Any	Allow
65500	DenyAllInBound	Any	Any	Any	Any	Deny
Outbound Security Rules						
65000	AllowVnetOutBound	Any	Any	VirtualNetwork	VirtualNetwork	Allow
65001	AllowInternetOutBound	Any	Any	Any	Internet	Allow
65500	DenyAllOutBound	Any	Any	Any	Any	Deny

Add or remove favorites by pressing Ctrl+Shift+F

Microsoft Azure

Search resources, services, and docs (G+/I)

Copilot

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Home > nsg-private

nsg-private | Subnets

Network security group

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Outbound security rules

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Subnets

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Locks

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Search subnets

Name	Address range	Virtual network
private-subnet	10.0.1.0/24	policy-lab

Give feed

Explanation:

In this step, a Network Security Group named nsg-private was created to control network traffic for the private subnet. The NSG was deployed in the East US 2 region under the policy-test-rg resource group. After creation, the NSG was associated with the private-subnet (10.0.1.0/24) within the policy-lab virtual network.

Step:5

Microsoft Azure

Search resources, services, and docs (G+/I)

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Home > nsg-private

nsg-private | Outbound security rules

Network security group

Search

Add

Hide default rules

Refresh

Delete

Give feedback

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Filter by name

Port == all

Protocol == all

Source == all

Destination == all

Action == all

Priority	Name	Port	Protocol	Source
100	Allow-Azure-Storage	8080	Any	Any
200	Deny-Internet	8080	Any	Any
65000	AllowVnetOutBound	Any	Any	VirtualNetwork
65001	AllowInternetOutBound	Any	Any	Any
65500	DenyAllOutBound	Any	Any	Any

Allow-Azure-Storage

nsg-private

Source

Any

Source port ranges

*

Destination

Service Tag

Destination service tag

Storage

Service

Custom

Destination port ranges

8080

Protocol

Any

TCP

UDP

ICMPv4

ICMPv6

Action

Allow

Deny

Priority

100

Save

Cancel

Give feed

Microsoft Azure

Search resources, services, and docs (G+/I)

Copilot

Home > nsg-private

nsg-private | Outbound security rules

Network security group

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Network security group security rules are evaluated by priority using the combination of source, source port, destination, destination port, and protocol to allow or deny the traffic. A security rule can't have the same priority and direction as an existing rule. You can't delete default security rules, but you can override them with rules that have a higher priority. [Learn more](#)

Filter by name

Port == all

Protocol == all

Source == all

Destination == all

Action == all

Priority	Name	Port	Protocol	Source
100	Allow-Azure-Storage	8080	Any	Any
200	Deny-Internet	8080	Any	Any
65000	AllowVnetOutBound	Any	Any	VirtualNetwork
65001	AllowInternetOutBound	Any	Any	Any
65500	DenyAllOutBound	Any	Any	Any

Deny-Internet

nsg-private

Source

Any

Source port ranges

*

Destination

Service tag

Destination service tag

Internet

Service

Custom

Destination port ranges

8080

Protocol

Any

TCP

UDP

ICMPv4

ICMPv6

Action

Allow

Deny

Priority

200

Save

Cancel

Microsoft Azure

Search resources, services, and docs (G+/I)

Copilot

Home > nsg-private

nsg-private | Outbound security rules

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Network security group security rules are evaluated by priority using the combination of source, source port, destination, destination port, and protocol to allow or deny the traffic. A security rule can't have the same priority and direction as an existing rule. You can't delete default security rules, but you can override them with rules that have a higher priority. [Learn more](#)

Filter by name

Port == all

Protocol == all

Source == all

Destination == all

Action == all

Priority	Name	Port	Protocol	Source	Destination	Action
100	Allow-Azure-Storage	8080	Any	Any	Storage	Allow
200	Deny-Internet	8080	Any	Any	Internet	Deny
65000	AllowVnetOutBound	Any	Any	VirtualNetwork	VirtualNetwork	Allow
65001	AllowInternetOutBound	Any	Any	Any	Internet	Allow
65500	DenyAllOutBound	Any	Any	Any	Any	Deny

Explanation:

In this step, outbound security rules were configured in the **nsg-private** Network Security Group to control external access from the private subnet. An **Allow-Azure-Storage** rule was created with priority **100** to permit outbound traffic to Azure Storage using a service tag. A second rule, **Deny-Internet**, was added with priority **200** to block all outbound internet traffic. These rules ensure that resources in the private subnet can access Azure Storage securely while preventing unrestricted internet access.

Step:6

Microsoft Azure

Search resources, services, and docs (G+/)

Copilot

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Home > Network foundation

Network foundation | Network security groups

Check security rule conflicts across network security groups | Suggest a firewall SKU for my network topology | Analyze traffic flow for network security groups

Search

Create | Manage view | Refresh | Export to CSV | Open query | Assign tags | Add to service group

Overview

Virtual network

Virtual Network overview

Virtual networks

NAT gateways

Public IP addresses

Network interfaces

Network security groups

Application security groups

Bastions

Route tables

Route servers

Private Link

DNS

Monitoring and management

You are viewing a new version of Browse experience. Click here to access the old experience.

Filter for any field...

Subscription equals all

Resource Group equals all

Location equals all

Add filter

Name	Resource Group	Location	Subscription	Flow log
nsg-private	policy-test-rg	East US 2	Azure for Students	
nsg-public	policy-test-rg	East US 2	Azure for Students	

Microsoft Azure

Search resources, services, and docs (G+/)

Copilot

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Home > Network foundation | Network security groups > nsg-public

Network found... | nsg-public | Subnets

Create | Manage view

You are viewing a new version of Browse experience. Click here to access the old experience.

Name

nsg-private

nsg-public

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Search

Associate

Search subnets

Name	Address range	Virtual network
public-subnet	10.0.2.0/24	policy-lab

Microsoft Azure

Search resources, services, and docs (G+/)

Copilot

pate1562@algonquin

Home > Network foundation | Network security groups > nsg-public

Network found... | nsg-public | Overview

Create | Manage view

You are viewing a new version of Browse experience. Click here to access the old experience.

Name

nsg-private

nsg-public

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How do I create an alert to track firewall metric failures | Analyze security rules for this network security group | Diagnose connectivity issues related to this security group

Move | Delete | Refresh | Give feedback

Essentials

Resource group : policy-test-rg

Location : East US 2

Subscription : Azure for Students

Subscription ID : 69be1f0a-e3f3-4038-96d6-3f6d028fd393

Tags : Add tags

Custom security rules : 1 inbound, 0 outbound

Associated with : 1 subnets, 0 network interfaces

Filter by name

Port == all

Protocol == all

Source == all

Destination == all

Action == all

Priority	Name	Port	Protocol	Source	Destination	Action
100	Allow-RDP	3389	TCP	Any	Any	Allow
65000	AllowVnetInBound	Any	Any	VirtualNetwork	VirtualNetwork	Allow
65001	AllowAzureLoadBalanc...	Any	Any	AzureLoadBalancer	Any	Allow
65500	DenyAllInBound	Any	Any	Any	Any	Deny
65000	AllowVnetOutBound	Any	Any	VirtualNetwork	VirtualNetwork	Allow
65001	AllowInternetOutBound	Any	Any	Any	Internet	Allow
65500	DenyAllOutBound	Any	Any	Any	Any	Deny

Explanation:

In this step, a Network Security Group named nsg-public was configured to allow remote access to resources in the public subnet. An inbound security rule Allow-RDP was created with priority 100 to permit TCP traffic on port 3389 from any source. This NSG was then associated with the public-subnet (10.0.2.0/24) in the policy-lab virtual network. This configuration enables secure Remote Desktop access to public-facing virtual machines while maintaining controlled network access.

Step:7

Microsoft Azure

Search resources, services, and docs (G+/I)

Copilot

Home > Storage center | Blob Storage >

Create a storage account ...

BasicsAdvancedNetworkingData protectionEncryptionTagsReview + create

Public access

Access your resource from anywhere through a public network.

Note: Allowing access to your resource through a public network increases security risk. [Learn more](#)

Public network access *

☒ Enable

Allow inbound and outbound access with the option to restrict select inbound access using resource access configurations for this resource.

☐ Disable

Restrict inbound access while allowing outbound access.

☐ Secure by perimeter (Most restricted)

Restrict inbound and outbound access using a network security perimeter. Secure by perimeter offers the greatest level of inbound and outbound restriction to secure your resource.

Public network access scope *

☐ Enable from all networks

☒ Enable from selected virtual networks and IP addresses

Virtual networks

Only the selected network will be able to access this storage account. [Learn more](#)

Virtual network subscription

Azure for Students

PreviousNextReview + create

Microsoft Azure

Search resources, services, and docs (G+/I)

Copilot

Home >

lab2storagetest

Storage account

Help me save costs on my storage. How can I make my storage account more resilient? How can I make this storage account more secure?

Search

UploadOpen in ExplorerDeleteMoveRefreshOpen in mobileCLI / PSFeedback

Overview

Activity logTagsDiagnose and solve problemsAccess Control (IAM)Data migrationEventsStorage browserStorage MoverPartner solutionsResource visualizerData storageContainersFile sharesQueuesTablesSecurity + networkingNetworkingFront Door and CDN

Essentials

Resource group : policy-test-rgLocation : eastus2Subscription : Azure for StudentsSubscription ID : 69be1f0a-ef3f-4038-96d6-3f6d028fd393Disk state : AvailableTags : Add tagsPerformance : StandardReplication : Locally-redundant storage (LRS)Account kind : StorageV2 (general purpose v2)Provisioning state : SucceededCreated : 1/20/2026, 8:02:59 PM

PropertiesMonitoringCapabilities (7)Recommendations (0)TutorialsTools + SDKs

Blob service

Hierarchical namespace : DisabledDefault access tier : HotBlob anonymous access : DisabledBlob soft delete : Enabled (7 days)Container soft delete : Enabled (7 days)Versioning : DisabledChange feed : DisabledNFS v3 : DisabledAllow cross-tenant replication : DisabledStorage tasks assignments : None

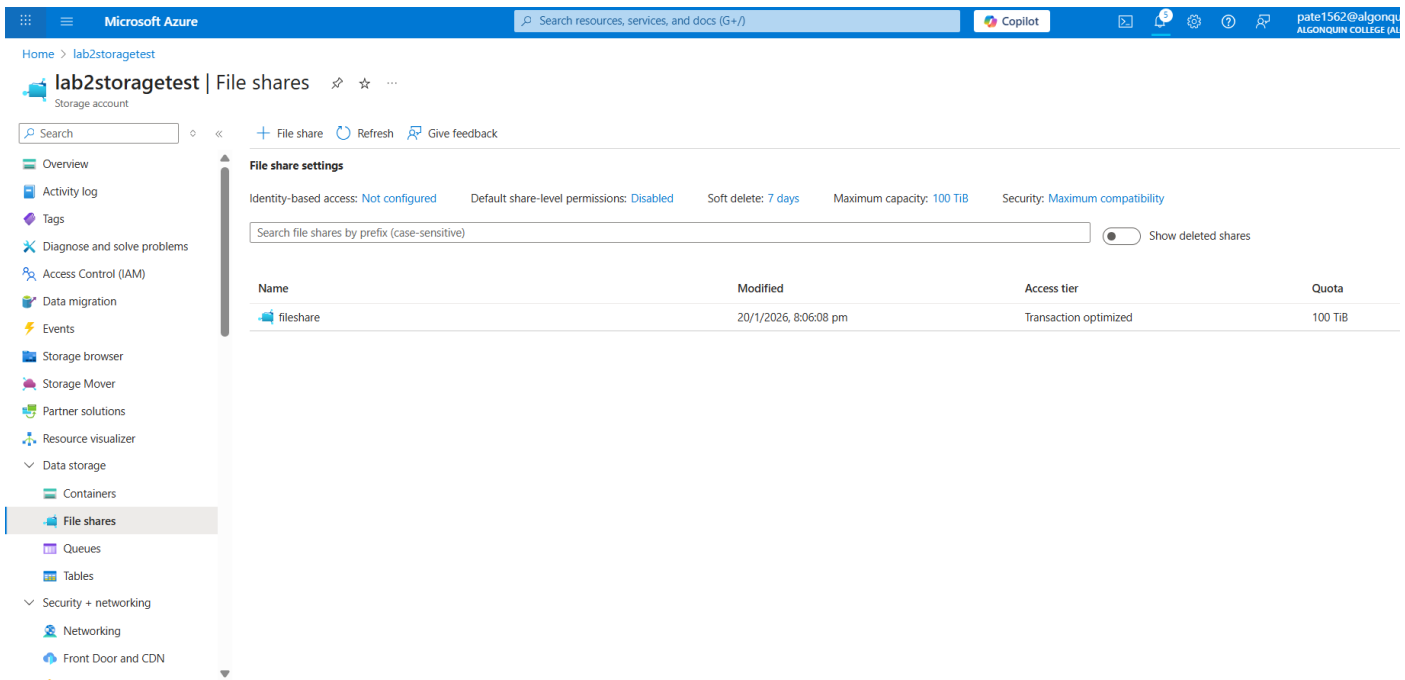
Security

Require secure transfer for REST API operations : EnabledStorage account key access : EnabledMinimum TLS version : Version 1.2Infrastructure encryption : Disabled

Networking

Public network access : EnabledPublic network access scope : Enable from selected virtual networks and IP addressesPrivate endpoint connections : 0Network routing : Microsoft network routingEndpoint type : Standard

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Explanation:

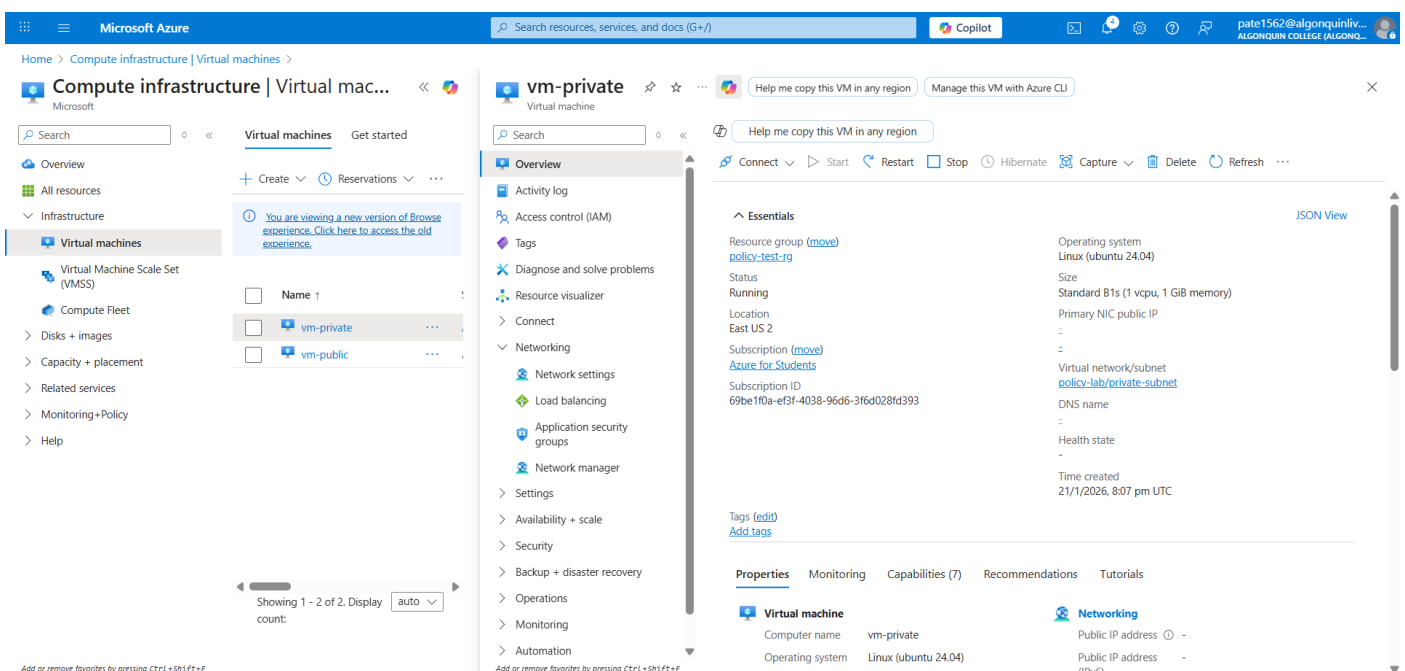
The storage account **lab2storagetest** was created with public network access enabled but restricted to selected virtual networks only.

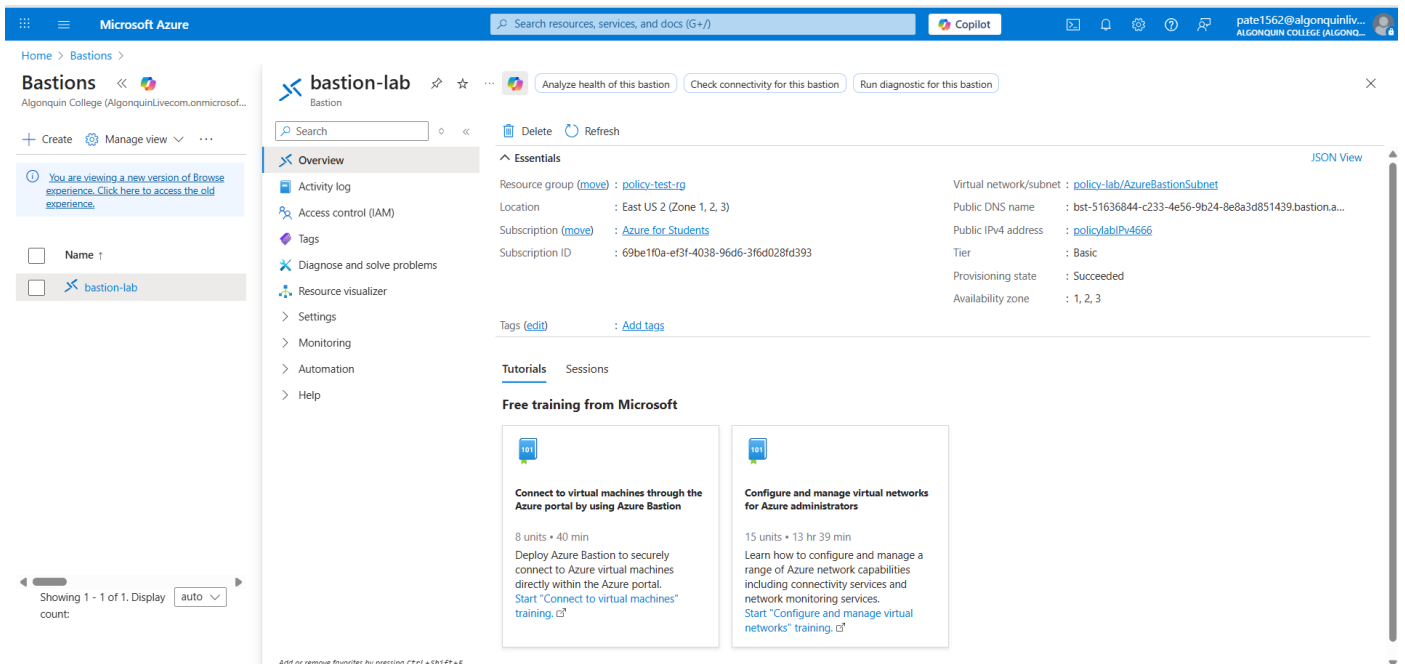
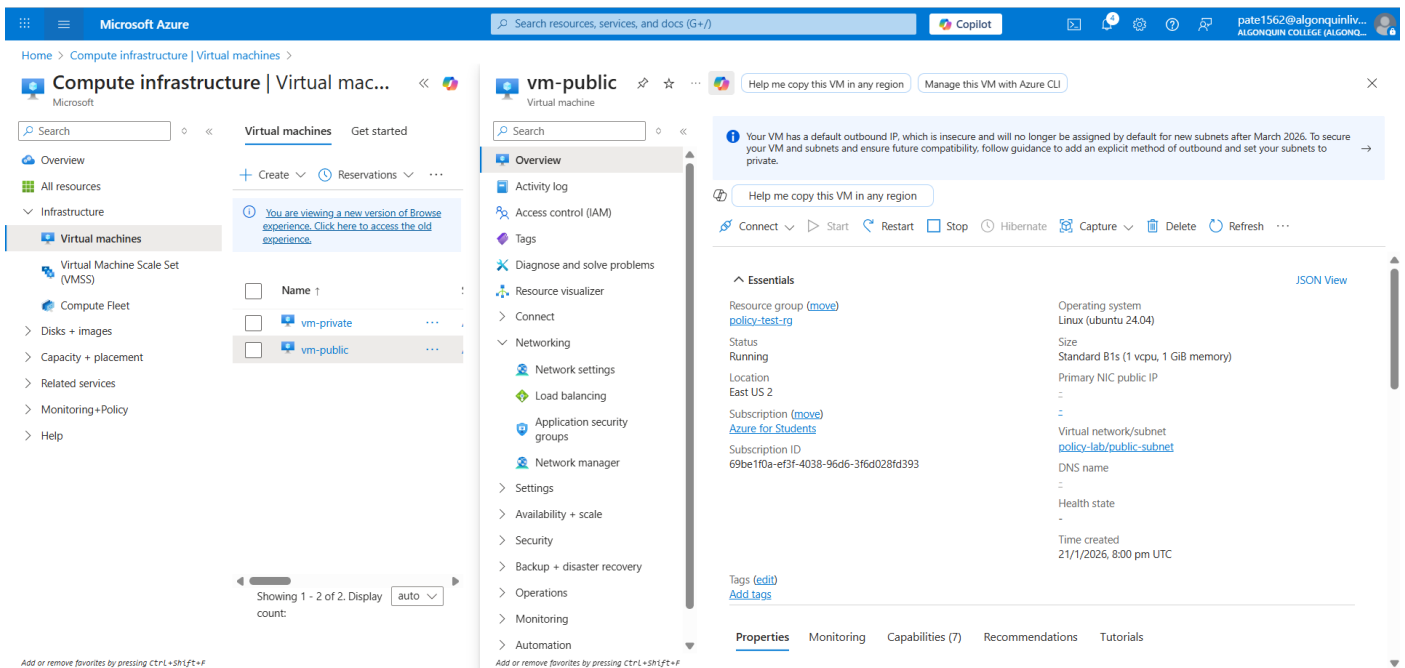
Access was explicitly allowed from the **private-subnet** within the **policy-lab** virtual network.

All other networks, including the public subnet, were automatically denied access.

An Azure File Share was successfully created and secured using these network restrictions.

Step:8





Explanation:

Two Linux virtual machines were created: **vm-private** in the private subnet and **vm-public** in the public subnet within the same virtual network.

Azure Bastion (**bastion-lab**) was deployed in a dedicated AzureBastionSubnet to allow secure browser-based access to both VMs.

This configuration ensures secure management access while keeping the private VM isolated from direct internet exposure.

Step:9

```
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-1017-azure x86_64)

* Documentation:  https://help.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:        https://ubuntu.com/pro

System information as of Thu Jan 22 00:02:25 UTC 2026

System load:  0.0          Processes:      116
Usage of /:   5.6% of 28.02GB Users logged in:  0
Memory usage: 30%         IPv4 address for eth0: 10.0.1.4
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status
>>

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

azureuser@vm-private:~$
```

```
azureuser@vm-private:~$ sudo apt install -y cifs-utils
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
cifs-utils is already the newest version (2:7.0-2ubuntu0.2).
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
azureuser@vm-private:~$ sudo mkdir /mnt/azurefiles
azureuser@vm-private:~$ sudo mount -t cifs //lab2storagetest.file.core.windows.net/filesshare /mnt/azurefiles \
> -o vers=3.0,username=lab2storagetest,password=IvjmYHmcL2zo256iLBmfGfT5BWiXLzEPMVyApK6BlguwB2y6rh0qLaa9oiMu0cIR9wmoPkJ3Zs00+AStxf0Yyw==
,dir_mode=0777,file_mode=0777,serverinoazureuser@vm-private:~$ df -h | grep azurefiles
//lab2storagetest.file.core.windows.net/filesshare 100T    0 100T   0% /mnt/azurefiles
azureuser@vm-private:~$ ls /mnt/azurefiles
azureuser@vm-private:~$ ^C
```

Explanation:

connected to **vm-private** using Bastion and mounted the Azure File Share using CIFS, which succeeded without errors.

This confirms that the storage account network rules are working correctly and **private-subnet access to Azure Files is allowed** as expected.

Step:10

```
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-1017-azure x86_64)
```

```
* Documentation:  https://help.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:       https://ubuntu.com/pro
```

```
System information as of Thu Jan 22 00:30:35 UTC 2026
```

```
System load:  0.03          Processes:            113
Usage of /:    5.7% of 28.02GB Users logged in:        0
Memory usage: 31%          IPv4 address for eth0: 10.0.2.4
Swap usage:   0%
```

```
Expanded Security Maintenance for Applications is not enabled.
```

```
0 updates can be applied immediately.
```

```
Enable ESM Apps to receive additional future security updates.
```

```
See https://ubuntu.com/esm or run: sudo pro status
```

```
The list of available updates is more than a week old.
```

```
To check for new updates run: sudo apt update
```

```
The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.
```

```
Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.
```

```
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.
```

```
azureuser@vm-public:~$
```

```
Expanded Security Maintenance for Applications is not enabled.
```

```
0 updates can be applied immediately.
```

```
Enable ESM Apps to receive additional future security updates.
```

```
See https://ubuntu.com/esm or run: sudo pro status
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Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.
```

```
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.
```

```
azureuser@vm-public:~$ sudo apt install -y cifs-utils
```

```
Reading package lists... Done
```

```
Building dependency tree... Done
```

```
Reading state information... Done
```

```
cifs-utils is already the newest version (2:7.0-2ubuntu0.2).
```

```
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
```

```
azureuser@vm-public:~$ sudo mkdir /mnt/azurefiles
```

```
azureuser@vm-public:~$ sudo mount -t cifs //lab2storagetest.file.core.windows.net/filesshare /mnt/azurefiles \
```

```
> -o vers=3.0,username=lab2storagetest,password=IvjYHmCL2zo256iLBmfGfT5BWIXLzEPMVyApK6BglguwB2y6rh0qLaa9oiMu0cIR9wmoPkJ3Zs00+AStxf0Yyw==,dir_mode=0777,file_mo
```

```
de=0777,serverino
```

```
mount error(13): Permission denied
```

```
Refer to the mount.cifs(8) manual page (e.g. man mount.cifs) and kernel log messages (dmesg)
```

```
azureuser@vm-public:~$ ^C
```

```
azureuser@vm-public:~$
```

Explanation:

This test confirms that the storage account is **not accessible from the public subnet** due to network restrictions. When the same Azure File Share mount command is run on **vm-public**, the request is blocked. The **“Permission denied”** error proves that access is correctly restricted to only the private subnet as intended.

Step:11

Microsoft Azure

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policy-test-rg
Resource group

Help me generate Terraform for this resource group configuration. How to manage these changes more efficiently with deployment tools? How to manage changes with deployment tools?

Search

Create Manage view Delete resource group Refresh Export to CSV Open query Assign tags Move Delete Export template

Overview

Activity log Access control (IAM) Tags Resource visualizer Events Settings Cost Management Monitoring Automation Help Support + Troubleshooting

Essentials

Resources Recommendations

Filter for any field... Type equals all Location equals all Add filter

Name	Type	Location
bastion-lab	Bastion	East US 2
lab2storagetest	Storage account	East US 2
nsg-private	Network security group	East US 2
nsg-public	Network security group	East US 2
policy-lab	Virtual network	East US 2
policy-lab-IPv4	Public IP address	East US 2
policylabIPv4666	Public IP address	East US 2
vault-mknbrgeo	Recovery Services vault	East US 2
vm-private	Virtual machine	East US 2
vm-private72	Network Interface	East US 2
vm-private_OsDisk_1_f83d67b68f52462897bbce283254c867	Disk	East US 2

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Resources Recommendations

Filter for any field... Type equals all Location equals all Add filter

Name	Type	Location
vault-mknbrgeo	Recovery Services vault	East US 2

Group by n